Observation of the antimatter hypernucleus ${}^{4}_{\Lambda}\overline{\mathrm{H}}$ nuclear lunch talk

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Paper overview and results

- The overarching goal of this paper is to use multiple antimatter asymmetry probes from QGP event expansion. This is a goal of the STAR experiment at RHIC.
- The probes used are the total sig/background count of the antimatter hypernucleus, lifetime probes of (anti)hypernuclei, and the ratio of hypernuclei matter and antimatter compared to previous experiment and theoretical predictions.

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Antimatter history

- Antimatter was first postulated by Paul Dirac in 1928 with negative energy solutions of the Dirac equation.

$$E = -\sqrt{p^2c^2 + m^2c^4}$$

- CPT symmetry (used in this paper) is a fundamental symmetry of physical laws, including charge conjugation, parity transformation, and time reversal.

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QGP formation, and Hyper-nuclei information

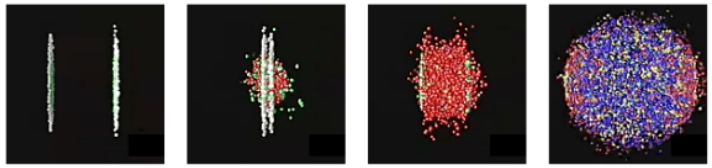
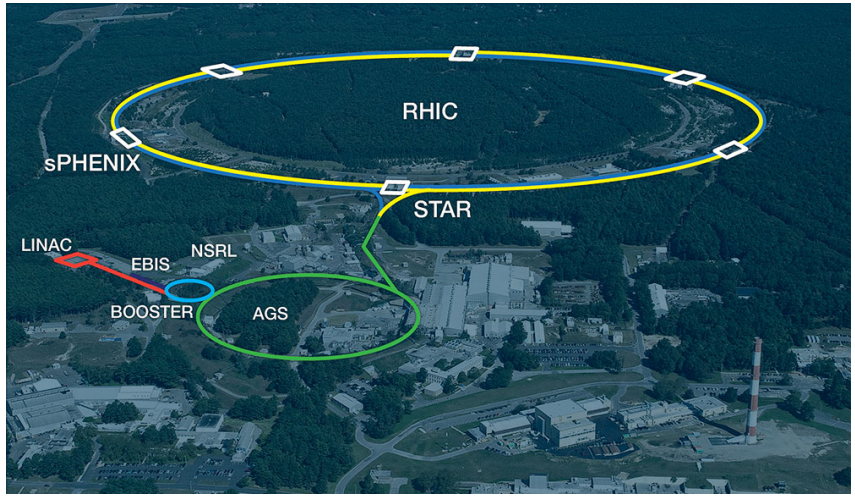


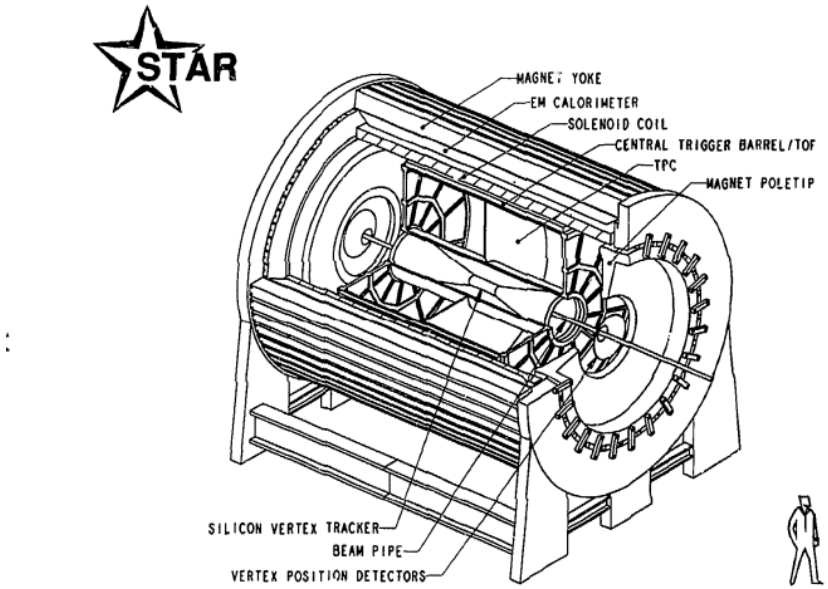
Figure 1.2: The leftmost panel shows two flattened gold nuclei about to collide. The left-center panel shows the initial collision. The right-center panel now shows freed partons. The rightmost panel shows the formation of the quark-gluon plasma. Image taken from [3].

- Quark-gluon plasma state creates conditions for matter/antimatter symmetry. Within the event expansion of QGP, antimatter can decouple within cooling stages, making it suitable to use heavy-ion collisions to generate antimatter. The above figure is an example of event expansion.
- The antimatter species used in this paper are hypertriton ($^{3}_{\Lambda}\overline{\mathrm{H}}$) and hyperhydrogen-4 ($^{4}_{\Lambda}\overline{\mathrm{H}}$).
- A hyperon is baryon type with one strange quark composite (lambda: uds). Hypernuclei are made from hyperons and nucleons.

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RHIC, STAR, and sub-detector systems (TPC and TOF) used





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RHIC, STAR, and sub-detector systems (TPC and TOF) used Cont.

- RHIC is a gigantic ring-shaped accelerator with a circumference of 3.8 km. It can accelerate heavy ions to 99.996 % of the speed of light. It can create collisions that deposit a lot of energy in small spaces, which make perfect conditions for QGP.
- STAR is one of the detectors used for QGP probes, spin structures of protons, and antimatter production.
- The STAR sub-detectors used mostly in this paper are the TPC and TOF for PID or particle tracking using energy loss per length or ionization.

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(Anti)hyper-nucleus reconstruction

- For reconstruction, the decay modes of the (anti)hypernuclei used were:

$$
\begin{aligned}
{}^{3}_{\Lambda}\mathrm{H} &\rightarrow {}^{3}\mathrm{He} + \pi^{-} \\
{}^{4}_{\Lambda}\mathrm{H} &\rightarrow {}^{4}\mathrm{He} + \pi^{-} \\
{}^{3}_{\overline{\Lambda}}\overline{\mathrm{H}} &\rightarrow {}^{3}\overline{\mathrm{He}} + \pi^{+} \\
{}^{4}_{\overline{\Lambda}}\overline{\mathrm{H}} &\rightarrow {}^{4}\overline{\mathrm{He}} + \pi^{+}
\end{aligned}
$$

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(Anti)hyper-nucleus reconstruction Cont.

- The daughter constituents are detectable based on their energy loss and ionization of the gas inside TPC.
- The main mechanism for this analysis for PID is the usage of the constituents' rigidity based on the effects that 0.5 T solenoidal has on energy loss ($< dE/dx >$).
- with the help of the TOF detector and other PID parts the squared mass over charge ratio is calculated.
- Equation 1 of the paper is an energy deviation measurement used per square mass charge ratio to isolate and get interested Helium-4 and Helium-3 matter and antimatter candidates.

$$n_\sigma = \frac{\ln\left(\frac{\langle dE/dx\rangle}{\langle dE/dx\rangle_{\text{th}}}\right)}{\sigma_{dE/dx}}$$

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PID Figures: 2a, 2b, and 2C

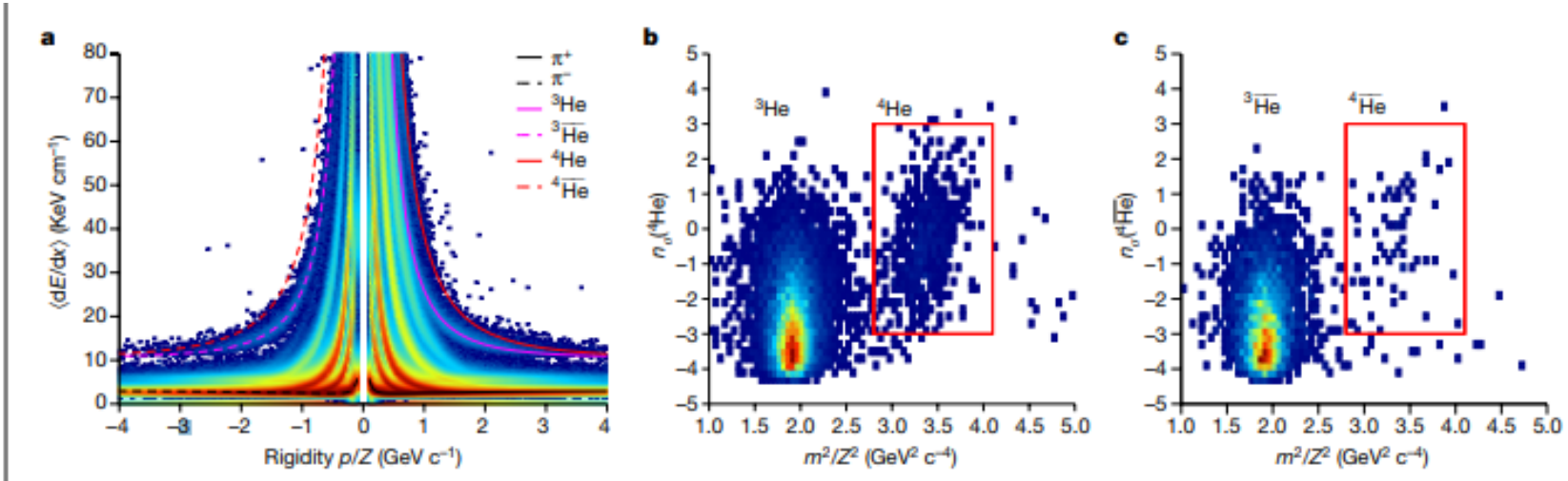


Fig. 2 | Particle identification based on energy loss and m^2/Z^2. a, Average energy loss $\langle dE/dx \rangle$ versus rigidity of charged particles measured by the TPC. The lines represent the expected trends for π^+, ^{3}He and ^{4}He and their corresponding antiparticles. **b**,**c**, $n_\sigma(^4\text{He})$ (**b**) and $n_\sigma(^4\overline{\text{He}})$ (**c**) versus m^2/Z^2. The red boxes indicate the region for ^{4}He and $^4\overline{\text{He}}$ candidates.

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Signals

- Each of the (anti)hypernuclei signals can be obtained by calculating the invariant mass of each of their daughter particles.
- the CM frame the Einstein mass energy relation and energy-momentum conservation. **Four-momentum:**

$$p^{\mu} = (E, \vec{p}) = (E, p_x, p_y, p_z)$$

Invariant mass squared (definition):

$$m^2 = p^{\mu} p_{\mu} = E^2 - |\vec{p}|^2$$

Invariant mass calculation for two particles:

$$m_{12}^2 = (p_1 + p_2)^{\mu}(p_1 + p_2)_{\mu} = (E_1 + E_2)^2 - |\vec{p}_1 + \vec{p}_2|^2$$

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Invariant mass plots

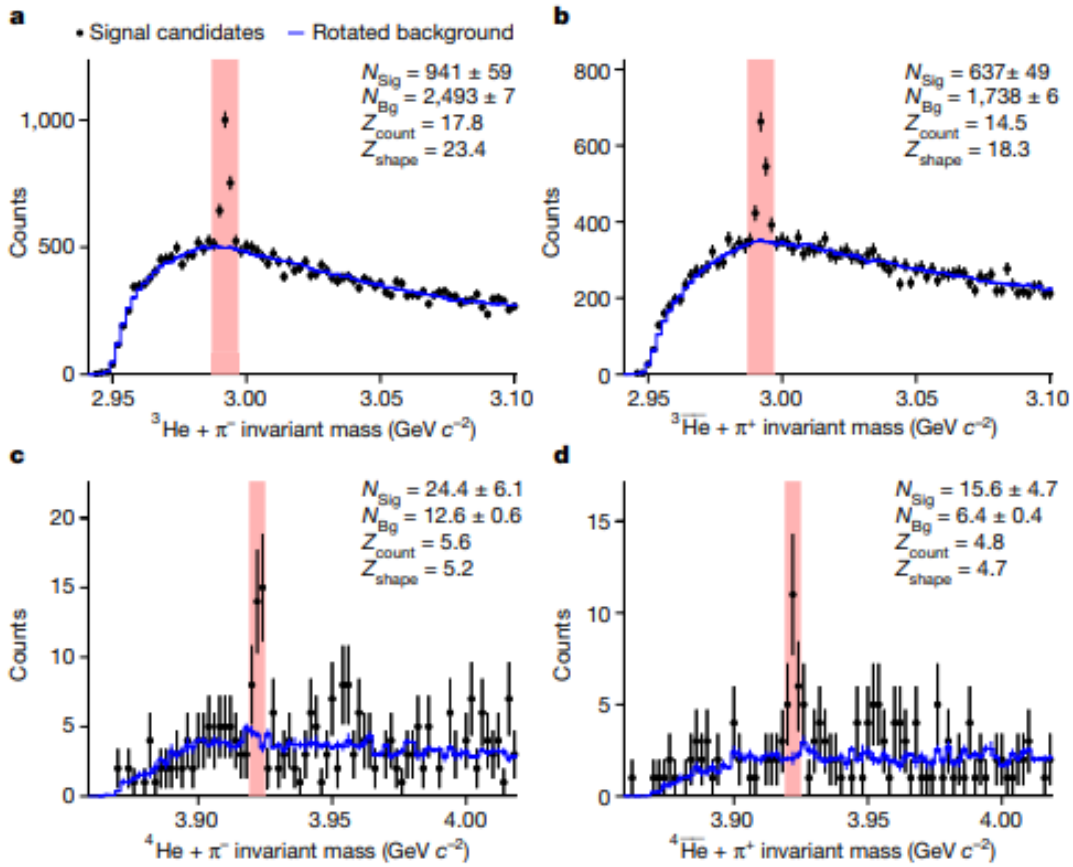


Fig. 3 | Invariant-mass distributions. a–d, Invariant-mass distributions of ^{3}He + π$^-$ (**a**), $^3\overline{\text{He}}$ + π$^+$ (**b**), ^{4}He + π$^-$ (**c**) and $^4\overline{\text{He}}$ + π$^+$ (**d**). The solid bands mark the signal invariant-mass regions. The obtained signal count (N_{Sig}), background count (N_{Bg}) and signal significances (Z_{count} and Z_{shape}) are shown in each panel.

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Results from Inv Mass plots

- the uncertainty probes for the counts used were asymptotic significance. Defined as equation 2. These numbers tell us if the effects observed by signal are because of actual physics or random flukes.

$$Z_{\text{count}} = \sqrt{2\left[(N_{\text{sig}} + N_{\text{bg}})\ln\left(1 + \frac{N_{\text{sig}}}{N_{\text{bg}}}\right) - N_{\text{sig}}\right]}$$

- The significances of the hypernuclei are 5.6 and 4.8 standard deviations. They correspond to p-values of 1.1×10^{-8} and 7.9×10^{-7}. The shape uncertainties corresponded to 5.2 and 4.7 standard deviations.

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Lifetime measurements

- To quickly restate the CPT theorem: physical laws remain the same under charge conjugation, parity transformation, and time reversal.
- Binding energies and lifetime probes are ways to investigate CPT symmetry limits.
- The paper has measured lifetimes of the hypernuclei and anti-hypernuclei species.
- It accomplishes said measurement by defining decay length $(L/\beta\gamma)$ bin ranges used for invariant mass distributions to obtain signal count for specific species. After applying efficiency corrections, the plot of yields per decay length bin is log-scaled and exponentially decay-fitted.

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Figure 4

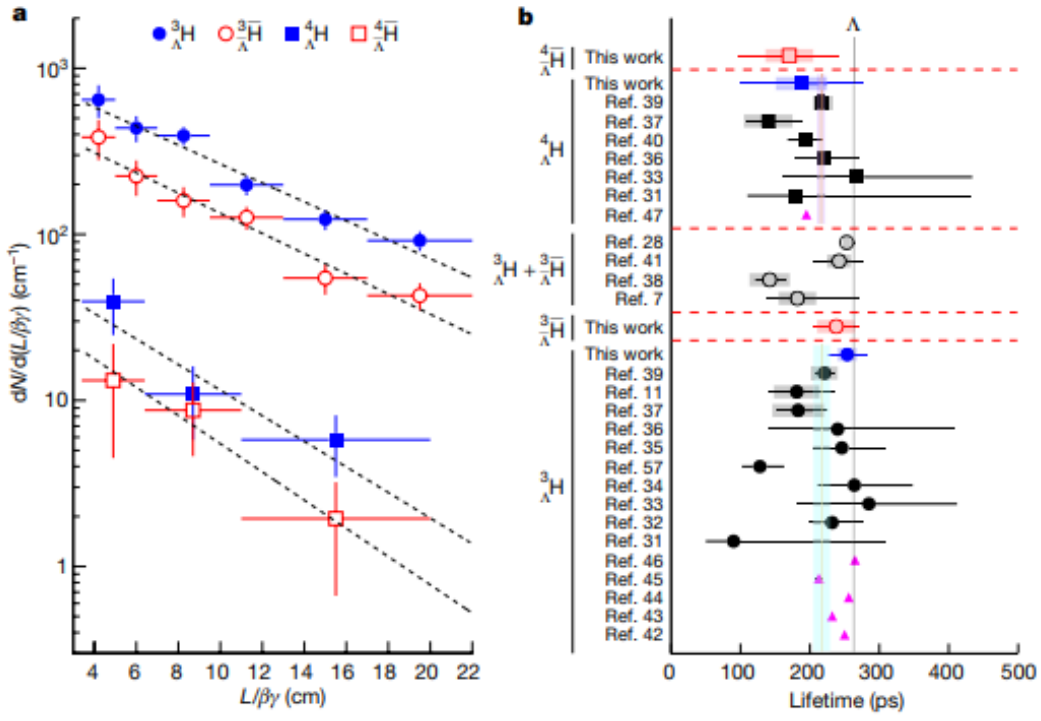


Fig. 4 | Hypernuclei dN/d($L/\beta\gamma$) as a function of $L/\beta\gamma$ and their lifetimes. **a**, $^{3}_{\Lambda}$H, $^{3}_{\Lambda}\overline{\text{H}}$, $^{4}_{\Lambda}$H and $^{4}_{\Lambda}\overline{\text{H}}$ yields versus $L/\beta\gamma$. The vertical error bars represent the statistical uncertainties only. **b**, Our measured $^{3}_{\Lambda}$H, $^{3}_{\Lambda}\overline{\text{H}}$, $^{4}_{\Lambda}$H and $^{4}_{\Lambda}\overline{\text{H}}$ lifetimes compared with world data[7,11,28,31–41,57] and theoretical predictions[42–47] (solid triangles). Error bars and boxes show statistical and systematic uncertainties, respectively. Solid vertical lines with shaded regions represent the average lifetimes of $^{3}_{\Lambda}$H and $^{4}_{\Lambda}$H and their corresponding uncertainties. These values are calculated from the previous results by a maximum-likelihood fit. The vertical grey line shows the lifetime of the free Λ.

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Lifetime results

- The experimental lifetimes extracted:

$$\tau\left({}^{3}_{\Lambda}\mathrm{H}\right) = 254 \pm 28(\text{stat.}) \pm 14(\text{sys.})\ \text{ps},$$
$$\tau\left({}^{3}_{\bar{\Lambda}}\overline{\mathrm{H}}\right) = 238 \pm 33(\text{stat.}) \pm 28(\text{sys.})\ \text{ps},$$
$$\tau\left({}^{4}_{\Lambda}\mathrm{H}\right) = 188 \pm 89(\text{stat.}) \pm 37(\text{sys.})\ \text{ps},$$
$$\tau\left({}^{4}_{\bar{\Lambda}}\overline{\mathrm{H}}\right) = 170 \pm 72(\text{stat.}) \pm 34(\text{sys.})\ \text{ps}.$$

- The lifetime differences for both hypernuclei:

$$\tau\left({}^{3}_{\Lambda}\mathrm{H}\right) - \tau\left({}^{3}_{\Lambda}\overline{\mathrm{H}}\right) = 16 \pm 43\,(\text{stat.}) \pm 20\,(\text{sys.})\ \text{ps}$$

$$\tau\left({}^{4}_{\Lambda}\mathrm{H}\right) - \tau\left({}^{4}_{\Lambda}\overline{\mathrm{H}}\right) = 18 \pm 115\,(\text{stat.}) \pm 46\,(\text{sys.})\ \text{ps}$$

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Yield Ratios

- production mechanisms of (anti)hypernuclei and (anti)nuclei can be inferred from heavy ion collisions.
- (anti)hypernuclei have binding energies of several Mev. This tells us that they are formed if they are in the last stage of QGP evolution. The coalescence model makes the production of (anti)hypernuclei highly unlikely.
- The goal of this stage of the paper is to use the coalescence model to get ratios of nuclei/anti-nuclei and anti-hypernuclei/hypernuclei to probe antimatter production.
- Three different collision systems were used to get these ratios.
- What was concluded with these results was that positive baryon production dominated the collision systems.

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Figure 5

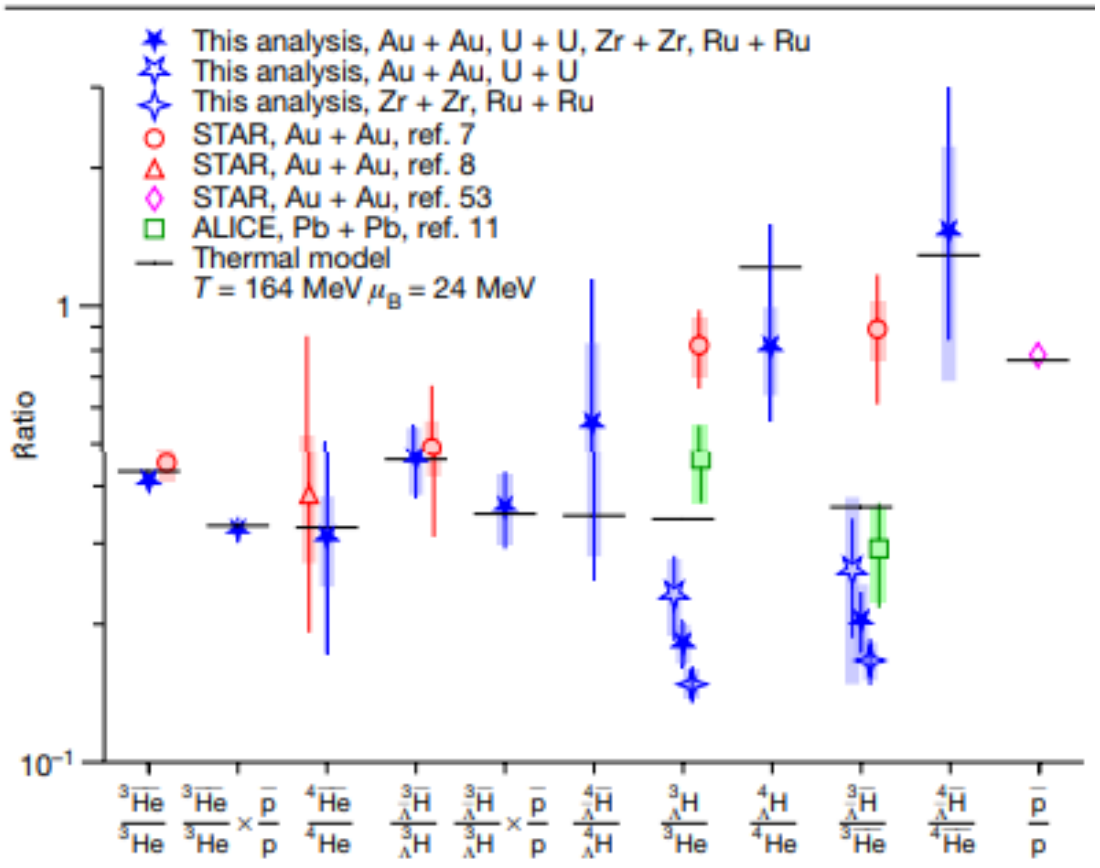


Fig. 5 | Production yield ratios among the various (anti)nuclei and (anti) hypernuclei with the same number of (anti)baryons. Results combining all collision systems in this work are shown by filled stars. Open stars show results with only U + U and Au + Au collisions, whereas quadrangular stars show results with only Zr + Zr and Ru + Ru collisions. Statistical uncertainties and systematic uncertainties are shown by vertical bars and shaded boxes, respectively. The decay branching fraction of ${}^{3}_{\Lambda}\mathrm{H} \to {}^{3}\mathrm{He} + \pi^-$ and ${}^{3}_{\Lambda}\overline{\mathrm{H}} \to {}^{3}\overline{\mathrm{He}} + \pi^+$ is assumed to be 0.25 (refs. 44,58), and the branching fraction of ${}^{4}_{\Lambda}\mathrm{H} \to {}^{4}\mathrm{He} + \pi^-$ and ${}^{4}_{\Lambda}\overline{\mathrm{H}} \to {}^{4}\overline{\mathrm{He}} + \pi^+$ is assumed to be 0.50 (refs. 59,60). Previous measurement results[7,8,11,53] and statistical-thermal-model predictions[48] are also shown for comparison.

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Conclusion

- This paper analyzed lifetimes, counts, and production ratios of (anti)hypertriton and (anti)hypernucleus.

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